Hongliang Li

+1 (814) 862-8126 | hjl5377@psu.edu | LinkedIn | Google Scholar | Website

RESEARCH PROFILE

Industrial Engineering Ph.D. candidate developing **optimization, control, and learning-enabled methods** for manufacturing and robotic systems. Experience in chance-constrained and bi-level optimization, supervised LLM fine-tuning, and robotic assembly algorithms, with evaluation through simulation and physical testbeds.

EDUCATION

The Pennsylvania State University — University Park, PA
Ph.D. in Industrial Engineering — Expected Dec 2026

Tianjin University — Tianjin, China
B.S. & M.S. in Civil Engineering — May 2017 & Jan 2020
Exchange: Nanyang Technological University, Singapore (2019–2020)

TECHNICAL SKILLS

Optimization & Control: LP/MIP; stochastic, chance-constrained, and bi-level optimization; model predictive control; strong duality; constraint generation; warm starts

Learning & Robotics: Supervised LLM fine-tuning; robot task planning; motion-sequence simplification; constrained point-cloud registration

Computing & Integration: Python, MATLAB, Gurobi, CVXPY, YALMIP, NumPy, Pandas, PyTorch, Git, Linux; OPC UA, PLCnext, Ignition

RESEARCH EXPERIENCE

Graduate Research Assistant — University Park, PA
The Pennsylvania State University — *Aug 2022 – Present*

Optimization and Control for Manufacturing Systems

- Develop **chance-constrained and bi-level optimization** models for shared-capacity allocation and production scheduling under uncertain demand, incorporating material flows, inventory limits, and service requirements.
- Implement **Python/Gurobi** models using primal-dual reformulations, strong duality, McCormick envelopes, selective constraint generation, and warm starts; examine conditions affecting formulation validity.
- Evaluate cost and service performance across **six demand-correlation levels**, using 20,000 fresh verification scenarios per round and 10,000 independent test scenarios per correlation level.
- Develop energy-aware production scheduling methods that reduced **energy cost by 34.2%** and combined energy and inventory holding cost by **2.9%** while meeting production requirements in manufacturing simulations.

LLM-Enabled Assembly and Geometric Optimization

- Fine-tune **GPT-4.1 mini** for robot primitive selection and structured command responses; achieved **100/100 correct responses versus 81/100** for the base model across explicit, multi-step, ambiguous, and safety-related command tests.
- Co-develop **motion-sequence simplification** for reusable assembly trajectories; reduced execution time by **81.7%** across nine component tasks versus recorded human-demonstrated trajectories in a fixed workspace.
- Co-develop **constrained ICP registration** using support-plane constraints and multiple pose initializations; reduced mean translation error from **5.74 to 0.36 mm** versus single-initialization ICP across eight assembly-board components in four simulated layouts.

ENGINEERING EXPERIENCE

Manufacturing Systems Engineering Intern — New Kensington, PA
Digital Foundry at New Kensington — *Jan 2026 – Present*

- Lead development of a **digital twin for an automated production line**, translating equipment dynamics, component flows, buffers, and process constraints into models; collaborate with controls and process engineers to refine assumptions.
- Validate predictions against a physical bottling line, achieving **6.8% production-time MAPE**; develop scheduling methods that reduced **energy cost by 6.1%** versus the benchmark while meeting all order deadlines.
- Integrate a **custom Python controller with a PLCnext PLC through OPC UA**; collect **5-Hz sensor and equipment data** in Ignition for parameter identification and closed-loop model validation.

Research Associate, Forecasting & Operations Analytics — Singapore
Surbana Jurong–NTU Corporate Lab — *Jan 2020 – Aug 2021*

- Developed Bayesian time-series and machine-learning models from noisy operational data to forecast system behavior and quantify uncertainty; translated findings into resource-allocation recommendations with engineering and operations stakeholders.

SELECTED PUBLICATIONS & MANUSCRIPTS

Li, H., Pangborn, H. C., & Kovalenko, I. (2026). Event-Triggered Runtime Planning for Coupled Decision-Making in Manufacturing Systems using Bi-level Optimization *IEEE Transactions on Automation Science and Engineering. Submitted.*

Li, H., Pangborn, H. C., & Kovalenko, I. (2026). Hierarchical Model Predictive Control for Energy-Aware Scheduling of Digital Twin-Based Batch Manufacturing Systems. *IEEE Transactions on Automation Science and Engineering.*

Li, H., Pangborn, H. C., & Kovalenko, I. (2026). Process-Informed Energy-Aware Batch Manufacturing Scheduling Using Hierarchical Digital Twins with a Bottling Plant Testbed. *ASME Journal of Dynamic Systems, Measurement, and Control.*

Li, H., Pangborn, H. C., & Kovalenko, I. (2026). Energy-Aware Integrated Proactive Maintenance Planning and Production Scheduling. *American Control Conference (ACC 2026).* Preprint: arXiv:2603.14060.

Li, H., Pangborn, H. C., & Kovalenko, I. (2026). Bi-level Model Predictive Control for Energy-Aware Integrated Product Pricing and Production Scheduling. *ASME Modeling, Estimation and Control Conference.* **Best Student Paper Finalist**.

Li, H., Pangborn, H. C., & Kovalenko, I. (2025, August). Energy-aware model predictive control for batch manufacturing system scheduling under different electricity pricing strategies. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 2096–2102). IEEE.

Li, H., Pangborn, H. C., & Kovalenko, I. (2023, August). A system-level energy-efficient digital twin framework for runtime control of batch manufacturing processes. In *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)* (pp. 1–6). IEEE.

Zhao, J., **Li, H.**, Lim, J., & Kovalenko, I. (2026). Speech-to-Trajectory: An LLM-Enabled Human-Robot Collaborative Framework for Generating New Assembly Tasks From Verbal Guidance. *IEEE International Conference on Automation Science and Engineering (CASE).*

Zhao, J., Lim, J., **Li, H.**, & Kovalenko, I. (2026). CoViLLM: An Adaptive Human-Robot Collaborative Assembly Framework Using Large Language Models. *arXiv preprint arXiv:2603.11461.*

Zhao, J., **Li, H.**, Lim, J., Hsiao, Y. N., & Kovalenko, I. Enabling Zero-Shot 6D Pose Estimation and Human-Assisted Classification for Flexible Manufacturing Assembly. *Manuscript.*

SELECTED RECOGNITION

ASME Dynamic Systems and Control Division Rising Star — 2025

MECC Best Student Paper Finalist — 2025, 2026

LANGUAGES & INTERESTS

Languages: Chinese, English

Interests: Tennis (USTA 4.0), golf, running, photography